WEEKLY UPDATE

Everything you asked for last week, answered.

Including one number that moved against us.

Suchith Rao & Tarun Senthil • for Dr. Carp

SINCE LAST WEEK

Four things you asked for

1	**Partition the variance**	How much does each factor explain?	**Done — table next slide**
2	**Re-check the 0.35**	Was stimulus removed properly?	**Done — the number dropped**
3	**Test for crosstalk**	Correlate cortex against muscle power	**Done — no contamination**
4	**Does it scale with learning?**	Failed animals as internal control	**Done — r = +0.64**

ASK 1 · THE PARTITION

What explains reflex size

Stimulus (M-wave)	47%	Dominant — how much stimulus reached the muscle
Background excitability	1%	Small alone, but a necessary control
Cortex, alone	21%	Mostly overlaps stimulus — stimulus drives both
Cortex, unique	2%	What cortex adds once the others are known
Unexplained	51%	Trial-to-trial variability we can't account for

Your read was right: the three don't add up, because cortex and stimulus share variance.

ASK 2 · THE CORRECTION

The 0.35 was inflated. The real number is 0.04.

What was wrong

I removed stimulus and background across ALL trials at once, then measured cortex within each 5-day block. If the stimulus–reflex relationship shifts between blocks, that leaves stimulus variance behind — and cortex tracks stimulus.

The fix

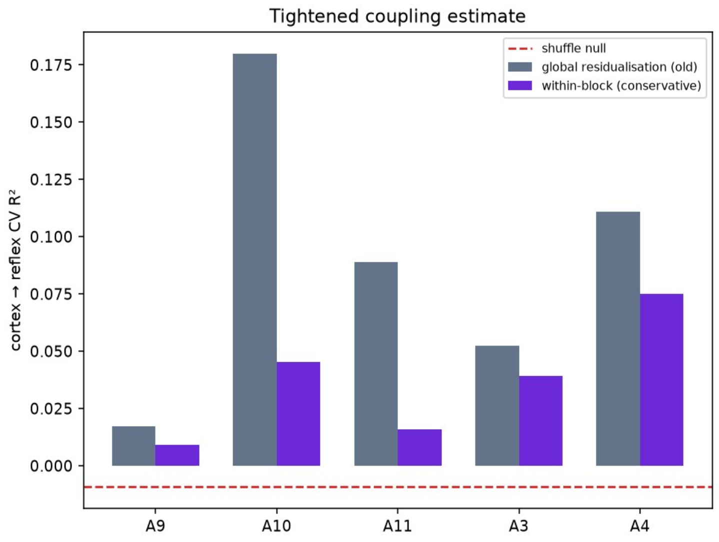
Remove them INSIDE each block, then decode. Nothing can leak across blocks. This is the conservative estimate and the one we'll quote.

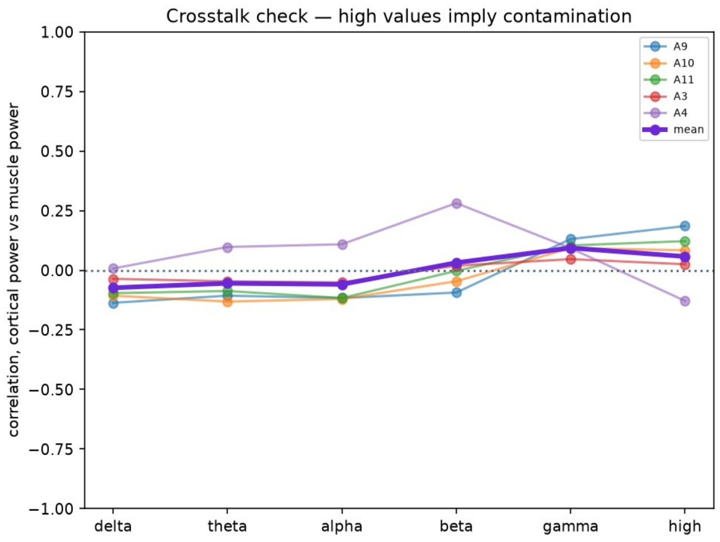
0.35	0.09	0.04	−0.01
best block, old method	mean, old method	mean, corrected	shuffle null

Still above the null in every animal — the finding holds, it's just smaller than I said.

ASK 3 · CROSSTALK

No muscle contamination





Your method

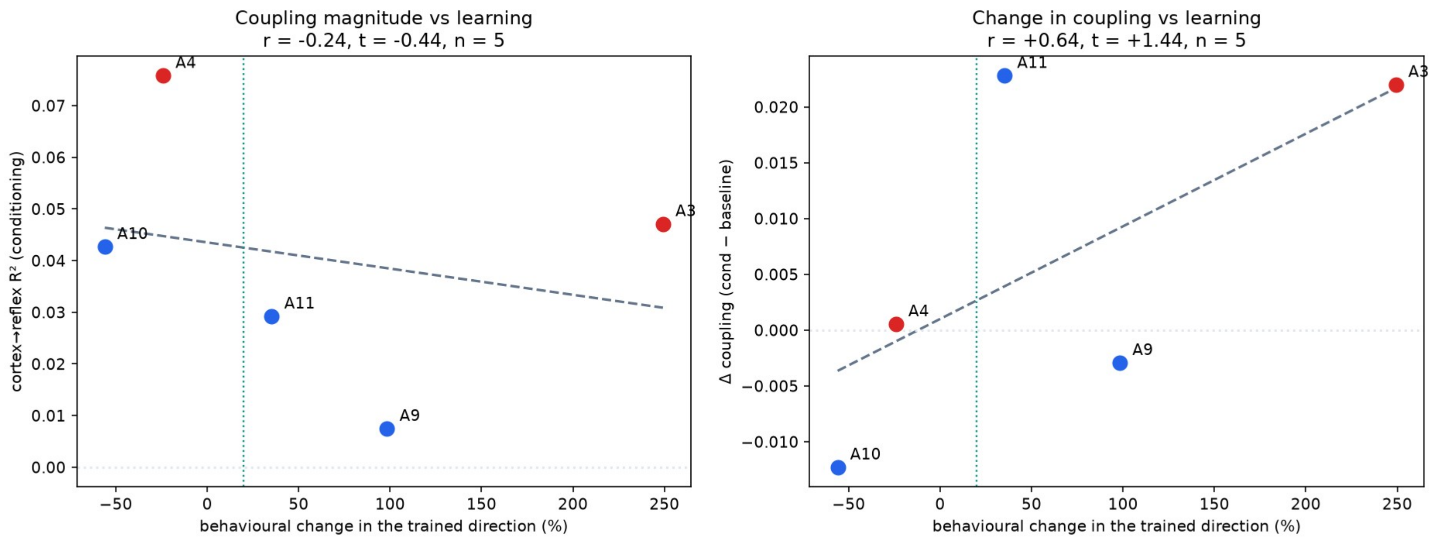
Correlate cortical power against muscle power, band by band. Muscle artefact lives at high frequency, so contamination would show there.

Above 100 Hz:
r = +0.06

Essentially zero. Low bands slightly negative. Cortex marginally leads muscle, which is the acceptable direction.

ASK 4 · THE INTERNAL CONTROL

The effect tracks how much they learned



Your idea

Animals that failed to condition shouldn't show the cortical effect. Two of ours failed — they're the built-in negative control.

$r = +0.64$

between learning and the change in coupling. The two successes gained coupling; the two failures didn't.

n = 5, so not significant yet.

NEW FINDING · A CONSTRAINT

We've run out of up-conditioned animals

Using the reward criterion as a success readout across every animal's log — the same trick that found animals 3 and 4:

A3	up	RW 90 → 190	bar raised — learned
A4	up	RW 150 → 220	bar raised — learned
A1	up	RW 110 → 59	bar lowered — struggled
A6	up	RW 150 → 80	bar lowered — struggled
A12	up	—	ECoG electrode dead (2.8 µV)

So the up group is capped at two. The down group can still grow — 7, 8, 13, 14, 15 are ready to download.

NEW THIS WEEK · DELIVERABLES

Everything now lives in one place

We put the project behind a single page so the code, the write-up and the slides stay in sync and are easy to hand to anyone.

REPOSITORY

The pipeline

Decoding, the pooled autoencoder, every control, per-animal results. One command per animal, reproducible end to end.

github.com/switchrao777/hroc-ncan

REPORT

Six-page write-up

Background, methods, results, limitations, references — with the four figures and the real numbers. Formatted as a handout.

docs/HROC_Report.pdf

SLIDES

The decks

This update, the talk for your group, and speaker notes for each.

slides/

Two blanks I left for you: Theresa's surname, and what Tarun and I should be listed as. I didn't want to guess on the write-up.

NEXT

Between now and the 21st

1. **Download and process the remaining animals** — 7, 8, 13, 14, 15 (down) plus 1 and 6 (failed up — negative controls). Takes n from 5 to about 12.
2. **Re-run everything at the larger n** — The scaling correlation is the analysis this powers. Fully automated — two commands per animal.
3. **Build the presentation for your group** — Pipeline explained with a data-flow figure, at the level you described.
4. **Preview it with you** — Before the 21st, so it's on par with what your group expects.

Question for you: with the up group capped at two, is the scaling analysis the right thing to build the paper on?